MD. SAKIB AL HASAN

Data Analyst | Data Scientist

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CAREER OBJECTIVE

Data Scientist with hands-on experience in Python, SQL, data analysis, machine learning, and data visualization. Built end-to-end projects involving customer segmentation, classification, EDA, feature engineering, and model evaluation. Seeking a Challenging Data Scientist or Data Analyst role.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn), SQL

Machine Learning: Linear & Logistic Regression, KNN, Naive Bayes, SVM, Decision Tree, Ensemble Methods (Random Forest, Gradient Boosting, AdaBoost, XGBoost), Neural Network (MLP), Clustering (K-Means, Hierarchical, DBSCAN), PCA

ML Workflow: Scaling & Encoding, Feature Engineering, Pipelines, Cross-Validation, GridSearchCV, RandomizedSearchCV, SMOTE

Model Evaluation: ROC-AUC, Precision, Recall, F1-Score, Confusion Matrix, RMSE, MAE, R^2, Silhouette Score, Elbow Method

Data Analysis & Excel: Data Cleaning, EDA, Data Visualization, Excel (PivotTable, Slicers, Power Query, Power Pivot/DAX, XLOOKUP)

Tools: Jupyter Notebook, Git, GitHub, AI-assisted development tools; MS Word, PowerPoint, Access

PROJECTS

1) Customer Segmentation using K-Means, Hierarchical & DBSCAN Clustering GitHub Repo

- Segmented **8,950 credit card customers** by spending and repayment behavior, engineering **15 features** via log-transforms, correlation filtering, and PCA (9 components, 95% variance).
- Benchmarked **K-Means, Hierarchical, and DBSCAN** (k=2–10) via silhouette score; selected **K-Means (k=2, silhouette 0.26)** as the best-performing, fully reproducible model in Python (pandas, scikit-learn).
- Identified **two actionable segments — 60% purchase-active customers vs. 40% cash-advance/high-balance** carriers — and translated each into concrete **marketing and credit-risk recommendations**.

2) Telco Customer Churn Prediction | Python, scikit-learn, XGBoost, SMOTE GitHub Repo

- Analyzed **7,043 telecom customers** and compared 6 ML models to predict churn, achieving **0.844 ROC-AUC** and **77% recall** on customers likely to leave.
- Found that **88.6% of churners** are on month-to-month contracts and recommended retention actions (auto-pay incentives, 1-year contract offers) for at-risk customers.

3) Medical Insurance Cost Prediction | Python, scikit-learn, XGBoost, SHAP GitHub Repo

- Built an end-to-end regression pipeline on **1,337 insurance records**; engineered smoker × BMI features that cut baseline cross-validated **RMSE by 25%**, then compared 8 models with 5-fold CV and hyperparameter tuning (XGBoost selected).
- Reached **R^2 of 0.85 (RMSE ≈ $4.6K)** averaged over 20 random splits; SHAP analysis showed **smoking status and age** drive most of the cost, with smokers averaging **~3.8x higher charges**.

4) Global Superstore Sales Dashboard | Excel, Power Query, DAX, VBA GitHub Repo

- Built a **4-page interactive Excel dashboard** on **25,035 retail orders ($12.6M sales, $1.47M profit)** using Power Query, Data Model, **DAX measures**, PivotCharts, slicers, and a VBA macro for one-click filter reset.
- Found that **discounts above 20% turn profit margin negative** and the top-selling product earns only a **6.8% margin**; recommended capping discounts at 20% and investigating low-margin markets (**Southeast Asia 2.0%, EMEA 5.4%**).

EDUCATION

B.Sc. (Hons) in Mathematics, National University — 4th Year (Running)
Govt. Shahid Asad College, Shibpur, Narsingdi | Session: 2021–2022 | Expected Completion: 2027

Higher Secondary Certificate (HSC) – Science
Narsingdi Public College, Narsingdi | 2021 | GPA 4.42/5.00

CERTIFICATIONS & TRAINING

- **Data Science & Machine Learning with Python** — Ostad 2026 | View Certificate
- **Computer Basic and ICT Applications** — Dept. of Youth Development, Ministry of Youth and Sports, Govt. of Bangladesh (Certificate ID: 141231, Grade: A+, Jul–Dec 2024); intermediate proficiency in MS Word, Excel, Access, and PowerPoint.